Weather, Sentiment, and the Stock Market

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Acknowledgment

This work was supported by the Ministry of Education of the Republic of Korea and the National Research Foundation of Korea(NRF-2021S1A5A2A03063960).

Preprint not peer reviewed

This preprint research paper has not been peer reviewed. Electronic copy available at: https://ssrn.com/abstract=4374950

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Abstract

This study examines the relationship between the weather and intraday investor sentiment. Our results indicate that high temperatures, high humidity, high cloud coverage, and extreme rain negatively affect investors' moods, whereas high winds and long sunshine durations improve their moods. Humidity, cloud coverage, and the sunshine duration significantly affect investor sentiment at market opening, whereas temperature, wind, and rain affect investor sentiment during the rest of the day. Stock market returns are weakly affected by cloud coverage but are unaffected by other weather variables. The weather has more prominent effects on turnover, volatility, and illiquidity. It has stronger effects on the market when accompanied by changes in investor sentiment. This study is the first to identify this intermediary through which weather affects market variables and to connect weather, sentiment, and markets.

Keywords: Intraday analysis; Investor sentiment; Stock market returns; Trading activity; Weather

JEL classification: G12, G14, G15

1. Introduction

Traditional finance theory assumes that all stock market investors are fundamentally rational and that the market is fully efficient. Investors construct well-diversified portfolios based on risk-return tradeoffs, meaning that unsystematic risk does not affect expected stock returns. In other words, according to traditional finance theory, market prices reflect only fundamental information about financial markets and are unaffected by investors' moods and irrational behavior. However, classical finance theory is limited in its ability to explain the movements of real financial markets, and over the past few decades, researchers have attempted to explain financial markets' abnormal movements from a behavioral finance perspective. Behavioral finance attempts to explain financial markets from a human rather than a rational perspective and proposes that people's psychological influences and cognitive biases, such as overconfidence, cognitive dissonance, and regret, affect investors' behavior and financial decision-making processes. As a result, psychological factors have a significant impact on the dynamics of asset prices.

One strand of the behavioral finance literature links to weather and financial markets. Psychology studies widely confirm that weather conditions, such as temperature, humidity, wind, and cloudiness, significantly impact individuals' moods (Allen & Fischer, 1978; Auliciems, 1972; Bitan, 1976; Bottema, 2000; Howarth & Hoffman, 1984; Hunt, Poulton, & Mumford. 1976; Page, Hajat, & Kovats, 2007; Penwarden, 1973; Redelmeier & Baxter, 2009; Schwarz & Clore, 1983; Toros, Deniz, Saylan, Sen, & Baloglu, 2005; Wyndham, 1969). Numerous psychological studies show that people evaluate future prospects more optimistically when they feel better (Arkes, Herren, & Isen, 1988; Bagozzi, Gopinath, & Nyer, 1999; Hirshleifer, 2001; Wright & Bower, 1992; Yuan, Zheng, & Zhu, 2006). According to the behavioral finance literature, investors' psychological biases, such as optimism or pessimism, affect financial markets; thus, the weather should affect financial markets.

Saunders (1993) is the first to identify a relationship between weather and stock markets. He analyzes the relationship between cloud cover in New York City and New York Stock Exchange (NYSE) index returns and finds that these returns are more likely to be negative on cloudy days. Hirshleifer and Shumway (2003) expand upon this study by examining the relationship between

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morning sunshine and market returns across 26 countries, and they find that sunshine is significantly correlated with market returns. Other studies extend this analysis by using additional weather variables. Kramer and Runde (1997) investigate the effects of cloud cover, humidity, and barometric pressure, and Keef and Roush (2007a) examine wind, temperature, rain, humidity, sunshine, and cloud cover. Capital markets in emerging countries are also significantly affected by the weather. Kang, Jiang, Lee, and Yoon (2010) analyze the effects of weather on the Shanghai Stock Exchange (SSE) and find that it affects returns and trading activities, such as turnover, volatility, and liquidity. Yoon and Kang (2009) and Shim, Kim, Kim, and Ryu (2015) investigate the Korea Composite Stock Price Index (KOSPI) market and identify a significant relationship between the weather and market variables, such as stock return volatility.

Although most of the aforementioned studies provide evidence of a relationship between weather and stock markets, they make a theoretical leap. Most previous studies directly connect market variables to the weather and investigate whether the market variables change when the weather does. However, the weather does not affect markets directly; instead, it indirectly affects markets through its effects on investor sentiment. That is, weather changes investor sentiment first, and these changes consequently affect markets. Thus, if the weather changes but investor sentiments remain unchanged for external reasons, the weather will not affect market variables. For example, even if the weather is very clear, if bad news is announced in the market, investors will become pessimistic. As a result, the market variables will move differently from the usual response to a lack of cloudiness. Thus, it is necessary to first analyze whether changes in weather variables impact investor sentiment. Nevertheless, previous studies do not perform this analysis because measuring investor sentiment in markets is challenging.

To overcome this limitation, we estimate intraday market sentiment in the KOSPI market and analyze the effects of weather on intraday market sentiment. We focus on the KOSPI market for three reasons. First, it has a high proportion of retail investors. As of 2022, the total trading volume in the KOSPI market is 146 billion shares, and the buying (selling) volume of retail investors is 106 billion shares (105 billion shares), accounting for 72.6% (71.9%) of the total volume. Retail investors are usually considered noise investors and are heavily influenced by psychological factors and sentiment (Berger, 2022; Cornelli, Goldreich, & Ljungqvist, 2006; Kumar & Lee, 2006; Kumar, Page, & Spalt, 2013). Thus, the KOSPI market is a suitable setting for studying the psychological effects on financial markets. Second, most financial institutions in Korea are concentrated in Yeouido, Seoul, and approximately half of Korea's population lives in the metropolitan area around Seoul.[1] Previous studies analyze the effects of the weather in New York and Shanghai on the NYSE and SSE, respectively. However, it is difficult to argue that the weather in New York or Shanghai affects the sentiment of all investors in the U.S. or China, respectively, because these countries' populations are widely distributed. Because Korea's population is highly concentrated around Seoul, the KOSPI market is suitable for analyzing the direct impact of the weather in Seoul on financial markets. Third, analyzing intraday weather effects in the KOSPI market can provide meaningful results. Chang, Chen, Chou, and Lin (2008) find that cloud cover has a significant effect on NYSE returns only when the market opens. Conversely, Lu and Chou (2012) find no effects of weather on the SSE market at the time of opening. They argue that this result arises because the trading systems in developed capital markets differ from those in emerging capital markets. The NYSE is a quote-driven market, whereas the SSE is an order-driven market; thus, most traders in the SSE do not need to reach their offices before the market opens. The effect of the weather on market opening is therefore less straightforward in the SSE. Instead, the

[1] As of 2021, Korea's total population is 51,738,071, and the population of the metropolitan area centered on Seoul is 23,124,656. Thus, the Seoul metropolitan area accounts for 44.70% of the total. In particular, the proportion of people in their 20s and 30s, who are more sensitive to psychological bias, is high at 48.52%.

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SSE breaks for lunch from 11:30 to 13:00, and the weather during this period significantly affects the SSE market at its afternoon session opening. Unlike the NYSE market, the KOSPI market is order-driven, and unlike the SSE market, it does not break for lunch. Thus, the effects of weather on intraday variables in the KOSPI market may differ from the results of previous studies.

For these reasons, we examine the relationship between the weather in Seoul and intraday sentiment in the KOSPI market. We also analyze the relationship between the weather and several intraday market variables, including returns, turnover, volatility, and illiquidity. We find that high temperatures, high humidity, high cloud coverage, and extreme rain negatively affect investors' moods. Conversely, high winds and long sunshine durations improve investors' moods. Humidity, cloud coverage, and sunshine significantly affect investor sentiment at market opening, whereas temperature, wind, and rain affect investor sentiment throughout the rest of the day. Market returns are weakly affected by cloud coverage but are not affected by other weather variables. However, turnover, volatility, and illiquidity are significantly affected by most of the weather variables. We also show that weather-induced changes in sentiment significantly affect market variables. In other words, we show that weather has a stronger effect on the market when it is accompanied by changes in investor sentiment. This study is the first to directly analyze the effect of weather on investor sentiment in the market. We directly analyze the intermediate paths through which weather affects market variables. This study is also the first to investigate the effect of weather on various intraday market variables in the KOSPI market and to connect weather, sentiment, and markets.

The remainder of this paper is organized as follows. In Section 2, we develop the hypotheses of the study, and in section 3, we describe our data and variables. Section 4 describes the methodology and empirical results, and Section 5 concludes.

2. Research backgrounds and hypotheses

Based on previous studies, we develop several hypotheses regarding the relationship between the weather, investor sentiment, and market variables. Previous studies explore the direct connections between weather and stock market variables. In other words, they directly investigate changes in market variables when weather conditions change. However, the weather does not directly affect financial markets; instead, it has an indirect effect through investor sentiment. In other words, weather affects a financial market via a two-step process. Weather first changes investor sentiment, and then the changes in investor sentiment affect the market. Previous studies do not analyze the first step because of limitations in measuring short-term sentiment. In this study, however, we estimate intraday market sentiment and analyze weather's effects on the estimated investor sentiment. Our first hypothesis is that weather significantly affects subsequent intraday investor sentiment.

Wyndham (1969) shows that individuals experience behavioral changes in the form of hysteria under extreme temperatures. Similarly, Cunningham (1979) argues that both high and low temperatures adversely affect people's moods. Page, Hajat, and Kovats (2007) also find that temperature has a negative effect on mood in the summer months. Thus, research has long confirmed that temperature significantly affects people's moods. In conjunction with financial markets, Chang, Nieh, Yang, and Yang (2006) show that temperature has asymmetric effects on the stock market; that is, extremely high and extremely low temperatures reduce subsequent market returns. Cao and Wei (2005) show that temperature has significantly negative effects on market returns in the U.S., Britain, Germany, Sweden, Japan, and Taiwan. In a recent study, Yan, Xiong, Li, and Lu (2022) reveal that a continuous rise in temperature affects stock returns significantly negatively. Thus, we expect to find that high temperatures reduce investor sentiment in the stock market.

Allen and Fischer (1978) and Auliciems (1972) show that individual performance decreases

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as humidity increases, and Howarth and Hoffman (1984) similarly find that people's moods worsen as humidity rises. In a financial context, Lu and Chou (2012) show that humidity negatively affects market returns in the period from 9:30 to 10:00, that is, the market opening period. Yoon and Kang (2009) also suggest that extremely high humidity has negative effects on stock returns from 1990 to 1997. Kang, Jiang, Lee, and Yoon (2010) and Shim, Kim, Kim, and Ryu (2015) identify a positive effect of humidity on implied volatility, suggesting a negative relationship between humidity and sentiment. Thus, we expect high humidity to reduce investor sentiment in the stock market.

Studies on environmental psychology show that strong winds cause discomfort, resulting in unpleasant moods (Bitan, 1976; Bottema, 2000; Hunt, Poulton, & Mumford, 1976; Penwarden, 1973). Toros, Deniz, Saylan, Sen, and Baloglu (2005) argue that wind speed has the most practical meteorological effect on human comfort, with high winds causing discomfort. In financial markets, Shu and Hung (2009) suggest that wind has a significantly negative effect on stock returns. Dowling and Lucey (2008) find a positive relationship between wind and variance in the equity market, indicating that wind negatively affects sentiment. Previous studies, therefore, suggest that wind and sentiment are inversely related.

The effects of cloudiness on mood have also been widely studied. Cunningham (1979) shows that sunshine causes people to experience good moods, and Schwarz and Clore (1983) suggest that people obtain more satisfaction from bright days than from gloomy days. More recently, Redelmeier and Baxter (2009) show that people interviewed on sunny days receive higher scores than those interviewed on rainy days do. Saunders (1993) is the first to connect cloudiness and financial markets and shows that market index returns are more likely to be negative on cloudy days. Numerous subsequent studies provide evidence of a negative relationship between the cloud cover ratio and market returns (Cao & Wei, 2005; Chang, Chen, Chou, & Lin, 2008; Chang, Nieh, Yang, & Yang, 2006; Keef & Roush, 2007b; Pardo & Valor, 2003). Thus, we expect to find that the cloud cover ratio negatively affects investor sentiment. If we measure cloudiness by the sunshine duration instead of by the cloud cover ratio, the sunshine duration should be positively associated with investor sentiment.

Our second question is whether the effect of weather on investor sentiment varies across different intraday trading intervals. Chang, Chen, Chou, and Lin (2008) analyze the effect of weather on market returns using NYSE data and show that the cloud cover ratio significantly reduces market returns only during the first 15 minutes after market opening. They argue that this result arises because the influence of information that is not related to the actual market, such as weather information, rapidly dissipates as more important information related to the financial market arrives over time once the market opens. Conversely, Lu and Chou (2012) use SSE data to show that the effect of the cloud cover ratio on index returns is significant in the trading period from 13:00 to 14:00. This period is immediately after the lunch break. The authors explain this result as follows. Unlike the NYSE, the SSE does not have market makers, meaning that most traders do not have to arrive at the office before the market opens. Thus, the weather has no significant effect on market returns during the market opening period. Instead, the SSE breaks for lunch from 11:30 to 13:00, and many traders leave the offices and return before the afternoon session. Thus, the weather during the lunch break significantly affects the SSE market from 13:00 to 14:00, during the afternoon session opens. Unlike the NYSE and the SSE, the KOSPI market has no market makers and no lunch breaks. Thus, it is unclear whether the effect of weather on the KOSPI market varies depending on the intraday trading interval, and we aim to answer this question.

Third, we investigate whether the relationships between weather and market variables identified in previous studies hold in the KOSPI market, which is one of the most developed emerging markets and which has a high proportion of noise investors. Specifically, we verify whether temperature, humidity, wind, and the cloud cover ratio have negative relationships with market returns and whether

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the sunshine duration has a positive relationship with market returns. We also analyze weather's effects on trading activity variables, including turnover, volatility, and illiquidity.

Finally, we determine whether sentiment amplifies weather's effect on market variables. As mentioned before, weather affects financial markets through the following two steps: first, it affects investor sentiment, and second, the changes in investor sentiment impact market variables. Thus, if the weather changes but investor sentiments do not change for other external reasons, then the market variables will not be affected by the weather. Thus, we analyze the effect of weather on market variables only during periods when the weather affects investor sentiment and check whether the effects of weather during this period are more significant.

3. Data and variables

3.1. Weather variables

We collect weather data for Seoul from Korea's National Climate Data Center, which provides various climatological statistics. Most of Korea's financial institutions are concentrated in Yeouido, Seoul, and more than half of its population lives in the metropolitan area centered in Seoul. Thus, we focus on the weather in Seoul. Among various available climate variables, we analyze the cloud cover ratio, humidity, sunshine duration, temperature, wind, and precipitation, which previous studies find to have significant effects on financial markets. We collect hourly observations of these climate variables between 9:00 and 16:00 from June 2000 to December 2021. Our time series begins in 2000 because Korean financial data are noisy and therefore less reliable before 2000. In addition, the Korean market closed from 12:00 to 13:00 until June 2000; thus, the intraday data for the market are discontinuous before June 2000. Thus, our sample starts in June 2000 and consists of 55,034 observations for each variable.

We calculate the apparent temperature (*TEMP*), which is the temperature equivalent that the human body perceives due to the motion of air, using the temperature and wind speed.[2] Humidity (*HUM*) is defined as the ratio between the water vapor pressure and the maximum amount of water vapor that can be included in the temperature at the observed time and is expressed as a percentage. *WIND* captures the wind speed at ten meters above the ground and is defined as the average wind speed for the ten minutes before the time of observation. The cloud cover ratio (*CC*) ranges from zero to ten and is defined as follows. The whole sky is set to ten, and the percent of the sky covered with clouds is assigned to the corresponding decile from zero to ten. The sunshine duration (*SS*) refers to the amount of time during which direct sunlight reaches the surface of the given geographical area without being blocked by clouds or fog and is expressed as a percentage of an hour. Finally, we define a dummy variable for rain (*RAIN*) that equals one if precipitation is positive during the observation period and zero otherwise.

Figure 1 shows the calendar-week average of each weather variable between 9:00 and 16:00 during the sample period. The left panel of Figure 1 shows the trends in *TEMP*, *CC*, and *SS*, and the right panel of Figure 1 shows the trends in *HUM* and *WIND*. Both panels clearly illustrate the seasonality of the weather variables. Jacobsen and Marquer (2008) argue that weather-related variables are highly correlated with the seasons; thus, it is difficult to distinguish between the effects of weather on financial markets and seasonal patterns in the markets. We therefore deseasonalize the raw weather

[2] Specifically, we calculate the apparent temperature (*TEMP*) using the following equation: $TEMP=13.12+0.6215T-11.37V^{0.16}+0.3965V^{0.16}T$. Here, *T* denotes the temperature in degrees Celsius, and *V* denotes the wind speed. This calculation was announced at the Joint Action Group for Temperature Indices conference in 2001, and it is the most widely used formula for apparent temperature in the world including Korea and North America.

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data. Specifically, we calculate the calendar-week mean for each weather variable and subtract the corresponding calendar-week mean from each observation. For example, we calculate the average value of *TEMP* for the 1,029 observations (21 years, seven days, and seven observations per day) in the first calendar week of the sample period.[3] We then subtract this mean from each hourly observation in the first week of each year. We denote these seasonally adjusted weather variables as $TEMP^d$, $WIND^d$, HUM^d, CC^d, and SS^d.

[Figure 1]

Saunders (1993) argues that days with 100% cloud cover are very different from other days, as such days are usually rainy. In other words, most partially cloudy days, including days with 90% cloud cover, have little or no rain; thus, weather-driven investor sentiment may not differ across different levels of partially cloudy days. We therefore define a dummy variable for extremely cloudy days (CC_{EXT}) that equals one if *CC* equals ten, that is, if the sky is fully covered with clouds, and zero otherwise. Similarly, we define a dummy variable for extreme sunshine (SS_{EXT}) that equals one if *SS* equals one, that is, if direct sunlight reaches the surface of the geographical area for the entire hour, and zero otherwise. Here, we use the raw values of *CC* and *SS* instead of their seasonally adjusted values, CC^d and SS^d. Finally, we construct an extreme rain dummy ($RAIN_{EXT}$) that equals one if the amount of precipitation falls in the top tertile.

3.2. Investor sentiment

We construct our intraday market sentiment variable following Seok, Cho, and Ryu (2021), who analyze the effects of firm-specific per-minute sentiment on subsequent stock returns. They first calculate the relative strength index (*RSI*), psychological line index (*PLI*), adjusted turnover ratio (*ATR*), and the logarithm of the trading volume (*LTV*) at every minute for each firm using trading data from the previous 15 minutes.[4] They then define firm-level sentiment as the first principal component obtained through principal component analysis (PCA). This component represents the optimal combination of these four proxies to maximize the overall variance. Using this sentiment index, Seok, Cho, and Ryu (2022) analyze the effect of macroeconomic news on intraday investor sentiment. In this study, we analyze the effect of weather on market-wide rather than firm-level sentiment. Thus, we calculate *RSI*, *PLI*, *ATR*, and *LTV* for the entire KOSPI market. To do so, we obtain per-minute KOSPI data from the Korea Exchange. The KOSPI market opens at 9:00 and closes at 15:30, but orders in the last ten minutes before the market close are made through synchronized bidding, meaning that no actual transactions take place between 15:20 and 15:30. Thus, our final dataset covers trades between 9:00 and 15:20 from June 2000 to December 2021.[5] We collect 1,930,994 per-minute KOSPI price observations and the total trading volume of the KOSPI market. Using these data, we calculate *RSI*,

[3] Our sample starts in June 2000. Thus, the sample period includes 21 first calendar weeks. We use 1,078 observations (22 years, seven days, and seven observations per day) to calculate the average for the 52nd calendar week.

[4] These four variables are widely used proxy variables for investor sentiment in previous studies (Baker & Stein, 2004; Chen, Chong, & Duan, 2010; Debata, Dash & Mahakud, 2018; Gao, Liang, Xu, & Xie, 2020; Gao & Yang, 2018; Hudson & Green, 2015; Hudson, Yan, & Zhang, 2020; Kim & Lee, 2022; Kim, Ryu, & Yang, 2021; Reis & Pinho, 2020; Ryu, Kim, & Yang, 2017; Ryu, Ryu, & Yang, 2020; Seok, Cho, & Ryu, 2019a; Yang & Zhou, 2016; Yao & Li, 2020). Seok, Cho, and Ryu (2019b) provide more details about each variable and explain why each variable can be used as a proxy for investor sentiment.

[5] On August 1, 2016, the closing time of the KOSPI market was changed from 15:00 to 15:30. Thus, for the period before August 1, 2016, our dataset includes trades made before 14:50.

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PLI, *ATR*, and *LTV* using the following equations.

$$RSI_t = \left[\frac{RS_t}{1+RS_t}\right], \text{ where } RS_{i,t} = \frac{\sum_{k=0}^{14} \max(P_{t-k} - P_{t-k-1}, 0)}{\sum_{k=0}^{14} \max(P_{t-k-1} - P_{t-k}, 0)} \quad (1)$$

$$PLI_t = \left[\sum_{k=0}^{14} \left\{\frac{\max(P_{t-k} - P_{t-k-1}, 0)}{P_{t-k} - P_{t-k-1}}\right\} / 15\right] \quad (2)$$

$$ATR_t = \sum_{k=0}^{14} \frac{15-k}{120} \times \frac{V_{t-k}}{S_{t-k}} \times \frac{R_{t-k}}{|R_{t-k}|} \quad (3)$$

$$LTV_t = \ln\left(\sum_{k=0}^{14} V_{t-k}\right) \quad (4)$$

Here, P_t denotes the price of the KOSPI at minute *t*, and V_t and S_t denote the total trading volume and total outstanding shares, respectively, in the KOSPI market at minute *t*. R_t is the log return over minute *t*, calculated as $R_t = ln\,(\frac{P_t}{P_{t-1}})$. We control for the volatility and momentum of the KOSPI to mitigate the common dependence of each proxy variable on short-term price movement pressures using the following equation.

$$PROXY_t = \alpha_0 + \beta_1 \times VOL_t + \beta_2 \times MOM_t + \varepsilon_t, \quad (5)$$

where $PROXY_t$ is one of the four sentiment proxies, that is, RSI_t, PLI_t, ATR_t, or LTV_t. VOL_t and MOM_t denote the short-term volatility and momentum, respectively, over the previous 15 minutes. We normalize each residual in Eq. (5) because each proxy has a different scale, and we denote these orthogonalized proxies as $RSI_t^{\perp}$, $PLI_t^{\perp}$, $ATR_t^{\perp}$, and $LTV_t^{\perp}$. Finally, we derive a composite sentiment index using the first principal component obtained from PCA. The detailed calculation is as follows.

$$S_t = F_{RSI} \times RSI_t^{\perp} + F_{PLI} \times PLI_t^{\perp} + F_{ATR} \times ATR_t^{\perp} + F_{LTV} \times LTV_t^{\perp} \quad (6)$$

where F_{RSI}, F_{PLI}, F_{ATR}, and F_{RSI} denote the first principal components of the proxies. In total, we construct 1,827,162 measures of per-minute market sentiment for 5,426 trading days. We calculate investor sentiment at hour *t* by averaging the overall per-minute sentiment from hour *t*-1 to hour *t*. For example, we define hourly investor sentiment at 10:00 as the average per-minute sentiment from 9:00 to 10:00 and match it with weather observations at 9:00.

3.3. Stock market variables

We calculate the returns, turnover, volatility, and illiquidity of the KOSPI on an hourly basis. The hourly returns of the KOSPI (*RET*) are calculated as logarithms such that $RET_t = ln\,(\frac{P_t}{P_{t-1}})$, where P_t is the closing price of the KOSPI at hour *t*. For opening-hour returns, we use the beginning price at 9:00 rather than the closing price on the previous day to exclude the influence of overnight returns. The last trading interval, from 15:00 to 15:20, is only one-third of an hour. We therefore match the returns in this period with the corresponding weather observations by multiplying them by three.[6]

Turnover (*TURN*) is defined as the total trading volume in the KOSPI market during each period divided by the total outstanding shares of all KOSPI firms. In the last trading interval, from 15:00 to 15:20, *TURN* is multiplied by three. Like Amihud (2002), we use the mean-adjusted value of turnover. In other words, we divide the turnover in each interval by its mean over the sample period to obtain

[6] Before August 1, 2016, the closing time of the KOSPI market is 15:00, and we use only the observations before 14:50. Thus, for this period, we multiply the returns during the last hour of the day by 6/5 to match the weather observations. We use the same process for turnover, volatility, and liquidity.

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TURN, and, thus, the mean of *TURN* is one.

We define volatility (*VOL*) as the difference between the highest and lowest prices in the given interval divided by the average of the two prices. That is, we calculate VOL_t as $(P_{(t-1,t),H} - P_{(t-1,t),L})/([P_{(t-1,t),H} + P_{(t-1,t),L}]/2)$, where $P_{(t-1,t),H}$ and $P_{(t-1,t),L}$ denote the highest and lowest prices of the KOSPI from time *t*-1 to time *t*. As with *TURN*, we multiply *VOL* by three for the last trading interval and use the mean-adjusted value of *VOL*.

The illiquidity (*ILLIQ*) measure is calculated following Amihud (2002). Specifically, $ILLIQ_t = \frac{|RET_t|}{VOLD_{(t-1,t)}}$, where $|RET_t|$ is the absolute value of the logarithm of the return in the given time period, and $VOLD_{(t-1,t)}$ is the dollar trading volume in Korean won from time *t*-1 to time *t*. As with *TURN*, we multiply *ILLIQ* by three for the last interval of the day, and we use its mean-adjusted value.

Table 1 presents descriptive statistics for the weather variables, sentiment, and market variables. Panel A shows statistics for the raw weather variables, Panel B shows statistics for the seasonally adjusted weather variables, and Panel C shows statistics for sentiment and market variables. Because we use the raw values of *RAIN*, *CC*, and *SS* to define the dummy variables *RAIN*, $RAIN_{EXT}$, CC_{EXT}, and SS_{EXT}, we do not present summary statistics for these variables in Panel B. On average, the apparent temperature of Seoul is 16.01°C, and the humidity is 52.96%. The average cloud coverage is 5.12, which means that half of the sky is covered with clouds on average. The average sunshine duration is 0.59, meaning that sunlight reaches Seoul for about 35 minutes out of each hour. However, the median sunshine duration is 0.80, indicating that the sun shines for at least 48 minutes per hour more than half of the year. Because we deseasonalize the weather variables by subtracting each calendar week's mean from the raw value, all seasonally adjusted variables shown in Panel B have a mean of zero.

[Table 1]

Panel C of Table 1 shows descriptive statistics for the sentiment and stock market variables. The hourly sentiment is 0.002 on average, with a median of 0.010, indicating that investors in the KOSPI market are optimistic on average. The average hourly return of the KOSPI is -0.001%, with a minimum return of -4.60% and a maximum return of 4.59%. The minimum return was recorded during the last trading interval (15:00 to 15:20) on March 26, 2020, when the COVID-19 shock occurred. Note that we multiply returns during the last trading interval by three to obtain an hourly return, so the actual return during that interval was -1.53%. The lowest return outside of the last trading interval is -4.03%, the return between 9:00 and 10:00 on October 23, 2008, which is the fourth lowest return in the sample. The maximum hourly return occurred between 13:00 and 14:00 on October 28, 2008. We use the mean-adjusted values of *TURN*, *VOL*, and *ILLIQ*; thus, their averages equal one. The minimum value of *TURN* is 0.13, indicating that the lowest turnover value is less than 87% of the mean. The maximum turnover value is almost 16 times the mean. Similarly, the lowest volatility is less than 99% of the mean, and the highest volatility is 17 times the mean. The minimum value of illiquidity is zero, which occurs when the hourly return is zero, and the highest illiquidity value is 22 times the mean. The minimum and Q1 values of *TURN*, *VOL*, and *ILLIQ* are all very different. Similarly, the differences between the maximum and Q3 values of these variables are quite large. These results show that the extreme values of the variables include many outliers.

4. Methodology and empirical results

4.1. Effects of weather on investor sentiment

To investigate the effects of weather on investor sentiment, we first conduct a simple sorting

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test. We match investor sentiment with the nearest previous observations of the weather variables. For example, we match the observation of $TEMP^d$ at 9:00 on June 1, 2000, with investor sentiment measured at 10:00, that is, the average per-minute sentiment from 9:00 to 10:00. We then divide the matched data into five groups according to the values of the weather variables and calculate the average sentiment for each group. Finally, we compare the average sentiment in the group with the highest weather values to that in the group with the lowest weather values. In the case of the dummy weather variables, that is, $RAIN$, $RAIN_{EXT}$, CC_{EXT}, and SS_{EXT}, we calculate the average sentiment when the variable equals zero and when it equals one and compare the two averages. This methodology is simple but intuitively shows the effects of weather on sentiment. However, the mean differences in the group may be driven by outlier values of sentiment in each group. Thus, we further define high sentiment as sentiment values above the median (0.01) and compare the percentages of high sentiment values within the highest and lowest weather groups. For example, if investor sentiment falls when $TEMP^d$ is low, we would expect average sentiment to be smaller in the lowest $TEMP^d$ group than in the highest $TEMP^d$ group. We would also expect to observe a greater percentage of high sentiment values in the highest *TEMP* group. The effects of the weather may vary with the seasons. For example, the temperature and humidity are high in the summer; thus, investor sentiment may decrease as $TEMP^d$ and HUM^d increase. However, because winter weather is cold and dry, $TEMP^d$ and HUM^d may have inverse effects on investor sentiment. Thus, we analyze the effects of the weather by defining summer as March to August and winter as October to February and considering both periods separately.

Table 2 presents the results of the sorting tests. For each weather variable, the first row, labeled *Average*, shows the results of comparing the average sentiment values, and the second row, labeled *Pr(>Med)*, shows the results of comparing the percentages of high sentiment values. The columns labeled *Low* and *High* show the average sentiment values (or percentages of high sentiment values) within the lowest and highest weather groups, respectively. Finally, the column labeled *Diff* shows the *t*-value (*z*-value) of the difference in the average sentiment values (percentages of high sentiment values) of the highest and lowest weather groups. We also consider summer and winter separately, and we report the *t*-values or *z*-values of the differences in the last two columns, labeled *Summer* and *Winter*.

[Table 2]

Table 2 shows that the average sentiment values for the highest and lowest $TEMP^d$ groups are -0.008 and 0.001, respectively, but the difference is not statistically significant. However, the percentage of high sentiment values in the highest $TEMP^d$ group is significantly smaller than the corresponding percentage in the lowest group, indicating that investor sentiment is more frequently below its median value when the temperature is higher. In other words, investors are more likely to be pessimistic as the temperature increases. This result can be explained by the separate results for summer and winter. In the summer, investor sentiment is significantly lower in the high-temperature group, and the percentage of high sentiment values is also low. Because we use seasonally adjusted temperatures, we can interpret this result as follows. In the summer, the average temperature is high, and if the temperature rises above the average, investors become depressed. Conversely, the temperature does not change investors' moods in the winter. Consequently, the overall effect of temperature on investor sentiment is negative.

The results for HUM^d and $WIND^d$ are similar to those for $TEMP^d$. Above-average humidity worsens investors' moods in the summer because the average humidity is already high. However, humidity has no significant effect in the winter. Similarly, the wind improves investors' moods during humid and hot summers but does not affect investors' moods in the winter. Thus, the overall effect of humidity is negative, and that of the wind is positive.

Table 2 also shows that a greater CC^d worsens investors' moods in the summer, but the difference is not significant. This result is inconsistent with previous studies showing that greater cloud

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cover lowers investors' moods, reducing market performance. According to our results, the sunshine duration has a greater effect on investor sentiment than the cloud cover ratio does. The longer the sunshine duration is, the more optimistic investors are, and this difference is significant. Thus, these results suggest that analyses of the weather's effects on financial markets should focus on the sunshine duration rather than the cloud cover ratio.

The last eight rows show the effects of extreme weather conditions on investor sentiment. Rain alone has no effect on investor sentiment, but extreme rain leads investors to become pessimistic, especially in the winter. As with the results for CC^d and SS^d, extreme clouds have no effect on sentiment, but extreme sunshine improves investors' moods.

Next, we conduct the same sorting test by dividing the trading day into the market opening period (9:00 to 10:00) and the rest of the day (10:00 to 15:20) to investigate whether the weather has a greater effect on investor sentiment immediately after the market opening. Table 3 shows the results. Here, we present only the average sentiment (percentage of high sentiment values) within the lowest-weather group (Low), that within the highest-weather group (High), and the *t*-value (*z*-value) of the difference between them. The first three columns of Table 3 show the results for the market opening period, and the last three columns show the results for the rest of the day. The results indicate that the effects of the weather at market opening do not grow stronger throughout the day. Instead, different weather variables affect sentiment at market opening and during the rest of the day. At the market opening, humidity, cloud coverage, the sunshine duration, and extreme sunshine affect investor sentiment, but none of these weather variables affects sentiment during the rest of the day. Instead, the temperature, wind, and rain change investors' moods throughout the rest of the day.

[Table 3]

The sorting test intuitively shows the effect of weather on sentiment, but it ignores key factors that affect the results. Thus, we regress investor sentiment on each weather variable. Our regression equation, given by Eq. (7), includes day-of-the-week dummies, month-of-the-year dummies, intraday trading interval dummies, and lags in sentiment as control variables.

$$S_t = \alpha + \beta_W W_{t-1} + \gamma_1 S_{t-1} + \gamma_2 D_t^{Mon} + \gamma_3 D_t^{Fri} + \gamma_4 D_t^{Jan} + \gamma_5 D_t^{Dec} + \sum_{k=1}^{6} \delta_k I_t^k + \varepsilon_t, \quad (7)$$

where S_t is the sentiment at time *t*, measured as the average of the overall per-minute sentiment from time *t*-1 to time *t*, and W_{t-1} is the weather variable at time *t*-1. We use the ranks of each variable instead of the raw sentiment and weather variables. In other words, we use rank regressions instead of simple ordinary least squares (OLS) because the scale of each weather variable is different and the scale of sentiment is different from that of each weather variable. For example, Table 1 shows that HUM^d ranges from -48.14 to 62.12, CC^d ranges from -8.16 to 6.88, and sentiment ranges from -6.48 to 8.08. Thus, it is difficult to compare the weather variables' effects on sentiment using simple OLS. Rank regressions also allow for nonlinearity in the relationship between weather and sentiment and eliminate the effects of outliers.

We include sentiment at time *t*-1 (S_{t-1}), also in rank form, to control for the persistency of sentiment, and we include dummy variables for Monday (D_t^{Mon}) and Friday (D_t^{Fri}) to control for day-of-the-week effects. We also control for month-of-the-year effects using dummy variables for January (D_t^{Jan}) and December (D_t^{Dec}). The intraday trading interval dummy variables, I_t^k, are defined as follows. I_t^1 equals one for the trading interval from 10:00 to 11:00 and zero otherwise, I_t^2 equals one for the trading interval from 11:00 to 12:00 and zero otherwise, I_t^3 equals one for the trading interval from 12:00 to 13:00 and zero otherwise, I_t^4 equals one for the trading interval from 13:00 to 14:00 and zero otherwise, I_t^5 equals one for the trading interval from 14:00 to 15:00 and zero otherwise, and I_t^6 equals

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one for the trading interval from 15:00 to 15:20 and zero otherwise. Thus, the baseline trading interval in each regression is the market opening period.

Hirshleifer and Shumway (2003) argue that a simple regression may not be appropriate for examining the effect of weather on market returns. They claim that although the sign of the market return is related to the weather, its magnitude may not be closely related to the weather. To account for this possibility, we estimate the logit regression presented in Eq. (8).

$$P(S_t > Median) = \frac{e^{\alpha+\beta_W W_{t-1}+\gamma_1 S_{t-1}+\gamma_2 D_t^{Mon}+\gamma_3 D_t^{Fri}+\gamma_4 D_t^{Jan}+\gamma_5 D_t^{Dec}+\sum_{k=1}^{6}\delta_k I_t^k}}{1+e^{\alpha+\beta_W W_{t-1}+\gamma_1 S_{t-1}+\gamma_2 D_t^{Mon}+\gamma_3 D_t^{Fri}+\gamma_4 D_t^{Jan}+\gamma_5 D_t^{Dec}+\sum_{k=1}^{6}\delta_k I_t^k}}, \quad (8)$$

where $P(S_t > Median)$ denotes the probability of a high sentiment value, that is, a value of sentiment above the median (0.01). In addition to estimating the regressions for the whole sample, we estimate each rank and logit regression separately for the market opening period and the rest of the day. Table 4 shows the results. We report only the coefficients of each weather variable owing to space limitations. The first three columns show the results of the rank regressions, and the last three columns show the results of the logit regressions. We report the *t*-statistics of the coefficients for the rank regressions and the chi-square statistics of the coefficients for the logit regressions. For each regression, the columns labeled *Open* show the results for the market opening period, and columns labeled *Other* show the results for the rest of the day.

[Table 4]

Table 4 shows similar results to those in Tables 2 and 3, indicating that the effect of weather on sentiment remains the same even after controlling for other factors. A high $TEMP^d$ reduces the probability of high sentiment and mainly affects sentiment after the market opening period. A high HUM^d also decreases investor sentiment but only in the market opening period. Both $WIND^d$ and SS^d have positive coefficients, but the former is only significant after the market opening period, and the latter is only significant during the market opening period. Under extreme rain, investors become more pessimistic, but not in the market opening period. We conduct the same analysis separately for summer and winter but do not present the results due to space limitations. The results are similar to those in Table 2.

In summary, this subsection shows that weather significantly changes investors' moods and that this effect is concentrated in the summer. The summer is hot and humid, meaning that above-average temperatures and humidity levels make investors more pessimistic. Conversely, summer winds and longer sunshine make investors optimistic. We also show that investors are more affected by sunshine than by cloud coverage. The only type of weather that affects investors' moods during the winter is extreme rain. We further show that the weather variables that affect sentiment differ during the market opening period and the rest of the day. Humidity, the cloud cover ratio, and the sunshine duration affect investors' moods in the market opening period but not during the rest of the day. The temperature, wind, and rain affect investors' moods throughout the rest of the day.

4.2. Effects of weather on market variables

In this subsection, we reconfirm the relationship between weather and financial market variables found in previous studies using intraday data. We first conduct a simple sorting test similar to that in Subsection 4.1 but use market variables instead of investor sentiment. For example, we match the weather variables observed at 9:00 with *RET* at 10:00, that is, the logarithm of returns from 9:00 to 10:00. We then divide the matched data into five groups according to the values of the weather variables

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and compare the average values of *RET* for the groups with the highest and lowest weather values. We also compare the percentages of the market variable observations above the median in the highest and lowest weather groups. Table 5 shows the *t*-values or *z*-values of the differences between the groups.

[Table 5]

Table 5 shows that among the various weather variables, only CC^d affects *RET*. In other words, the higher the cloud coverage is relative to the seasonal average, the lower the probability of higher returns is. Chang, Chen, Chou, and Lin (2008) argue that wind, rain, and temperature have little effect on market returns, and Yoon and Kang (2009) similarly find no significant effects of humidity and temperature on the KOSPI market. Lu and Chou (2012) show that temperature, humidity, wind, and rain do not influence subsequent index returns. Pardo and Valor (2003) and Kang, Jiang, Lee, and Yoon (2010) also report that humidity does not affect market returns. Conversely, numerous existing studies identify a negative effect of cloud cover on stock returns. Saunders (1993) demonstrates that market returns tend to be lower in the U.S. when the weather is cloudy. Hirshleifer and Shumway (2003) show that cloudiness is negatively correlated with stock market index returns across 26 countries. Thus, the results in Table 5 are largely consistent with those of previous studies.

Whereas *RET* is not affected by most of the weather variables, Table 5 shows that the trading activity variables, that is, *TURN*, *VOL*, and *ILLIQ*, significantly differ depending on the weather. These results are consistent with those of Lu and Chou (2012), who show that although no weather variables affect returns, the temperature, wind, humidity, and cloud coverage have significant effects on other market variables. Chang, Chen, Chou, and Lin (2008) also show that although the cloud cover ratio, wind, and temperature do not affect index returns, they have significant effects on volatility, market depth, and turnover. Consistent with these studies, we show that increases in $TEMP^d$ and $WIND^d$ reduce *TURN*, *VOL*, and *ILLIQ* and that a rise in HUM^d increases these variables. In addition, we find that SS^d only affects *VOL* and *ILLIQ* and that CC^d only affects *RET*. However, extreme clouds and rain increase all three trading activity variables.

Next, we regress each market variable on each weather variable. Our regressions include lags in the market variables, day-of-the-week dummies, month-of-the-year dummies, and intraday trading interval dummies as control variables. We replace sentiment (S_t) in Eq. (7) with each market variable and estimate the regression 36 times (nine weather variables for each of the four market variables). We use the ranks of each weather and market variable instead of the raw values, as in the regressions for the sentiment. Table 6 reports the coefficient of each weather variable and its *t*-statistic.

[Table 6]

Table 6 shows similar results to those in Table 5. $TEMP^d$, HUM^d, $WIND^d$, and SS^d have little effect on *RET*, whereas CC^d has a significantly negative effect on *RET*. When we include other control variables, $RAIN^d$ has a positive effect on RET. Although we do not report the results for brevity, when we divide the entire trading interval into the market opening period and the rest of the day, rain has a negative effect during the market opening period and a positive effect throughout the rest of the day. Table 6 also confirms that $TEMP^d$, $WIND^d$, and SS^d have negative effects on trading activity and that HUM^d has a positive effect on trading activity. CC^d has no effect on the three trading activity variables, but extreme cloudiness increases *TURN*, *VOL,* and *ILLIQ*. The results of the logit analysis of the probability that each market variable is above its median are similar to those of the rank regressions.

In summary, we confirm the expected relationships between the weather and financial market variables and show that only cloud coverage affects market returns. We find that most of the weather variables, except for cloud coverage, have significant effects on the trading activity variables.

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4.3. Weather, sentiment, and the stock market

Now, we link the results in Subsections 4.1 and 4.2 to analyze the intermediate path through which weather affects market variables. Previous studies on the effect of the weather on stock markets directly connect the weather to market variables. They directly investigate the changes in market variables according to the weather. However, the weather affects the market via a two-step process. First, the weather affects investor sentiment. Second, the changes in investor sentiment affect the market variables. Thus, if the weather changes but investor sentiments do not change for other, external reasons, the market variables will not be affected. For example, even if cloud coverage is low, if bad news is announced, investors will become pessimistic. As a result, the market variables will move differently from the usual response to the low cloud coverage. In this subsection, we identify the effects of weather on market variables only when investor sentiment is affected by the weather.

First, we briefly analyze the simultaneous effects of intraday sentiment on the market variables. Analyses of investor sentiment and markets typically analyze the effects of sentiment on subsequent market variables; this study, however, investigates the effect of the weather on subsequent investor sentiment and the simultaneous changes in the market variables based on the changes in investor sentiment. Thus, we analyze the simultaneous effects of sentiment on the market variables. To do so, we match investor sentiment at time t with the corresponding market variable at time t and divide the full sample into two groups: periods with positive sentiment values and periods with negative sentiment values. We then compare the averages of *RET*, *TURN*, *VOL*, and *ILLIQ* in each group. The results are presented in Table 7.

[Table 7]

Table 7 shows that periods with high sentiment have significantly higher returns than periods with low sentiment, indicating that intraday sentiment has a positive effect on short-term returns. This result is consistent with Seok, Cho, and Ryu's (2021) finding that intraday investor sentiment is positively related to subsequent short-term returns. Similarly, Edmans, Fernandez-Perez, Garel, and Indriawan (2022) and Fang, Chung, Lu, Lee, and Wang (2021) suggest that sentiment is positively related to market returns in the same week or even in the same month. Conversely, sentiment is negatively correlated with the other trading activity variables, that is *TURN*, *VOL*, and *ILLIQ*. Baker and Stein (2004) and Lee, Jiang, and Indro (2002) argue that investor sentiment is negatively related to market volatility because when investors are pessimistic, they disagree more about the market. Liu (2015) demonstrates that when investors are more optimistic, the stock market is more liquid, meaning that illiquidity is negatively associated with investor sentiment. Thus, the results in Table 7 are in line with those of previous studies.

Finally, we connect the weather, sentiment, and market variables. As mentioned at the beginning of this subsection, weather that does not affect investor sentiment does not affect the market. Thus, we perform the analysis in Subsection 4.2 again for only the period in which the weather actually affects investor sentiment in the expected direction. For example, in Subsection 4.1, we show that investor sentiment falls when the temperature is high; thus, we can infer that other factors are affecting sentiment if a period of high sentiment follows a high-temperature recording. We define a period of low sentiment following a high temperature as a *True* period because the weather affected actual investor sentiment in the expected direction. Similarly, a period of high sentiment following a low temperature is defined as a *True* period. According to this classification, periods of high (low) sentiment following low (high) values of $TEMP^d$, HUM^d, and CC^d are *True* periods. Periods of high (low) sentiment following high (low) values of $WIND^d$ and SS^d are also *True* periods. For each weather variable, we divide the sample period into five groups according to the weather values and then separate each group into two subgroups according to the sign of sentiment. We retain only the *True* periods and conduct a sorting test similar to that in Table 5. Here, we compare only the averages of market variables and not

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the percentages of values above the median. We do not analyze the weather dummy variables (i.e., $RAIN$, $RAIN_{EXT}$, CC_{EXT}, and SS_{EXT}). Table 8 shows the results.

[Table 8]

In contrast to the results in Table 5, all the weather variables have significant effects on market returns in Table 8. The results suggest that low market returns follow high $TEMP^d$, HUM^d, and CC^d values, whereas high market returns follow high $WIND^d$ and SS^d values. Combining the results in Table 8 with those in Tables 2 and 7, we can draw the following conclusions. High temperatures, humidity levels, and cloud cover ratios lower investor sentiment, and this decreased investor sentiment induces low market returns. Similarly, high winds and sunshine durations increase investor sentiment, which induces high market returns. The trading activity variables follow the same mechanism, but the sentiment is inversely related to these variables, as Table 7 shows. For example, higher humidity and cloud cover reduce investor sentiment and, thus, increase *TURN*, *VOL*, and *ILLIQ*. Conversely, greater wind and sunshine duration increase investor sentiment and, thus, decrease *TURN*, *VOL*, and *ILLIQ*. The only weather variable that does not fit this pattern is temperature. Tables 5 and 6 show that as the temperature increases, *TURN*, *VOL*, and *ILLIQ* decrease significantly; however, according to Table 8, a change in temperature accompanied by a change in investor sentiment does not affect the trading activity variables.

We also conduct a regression analysis similar to that in Table 6 for only the *True* periods described previously. Table 9 presents the coefficients of each weather variable and their significance levels. The results are almost the same as those in Table 8 except that CC^d is no longer significant. $TEMP^d$, HUM^d, and CC^d have negative coefficients, indicating that these weather variables reduce market returns when accompanied by changes in investor sentiment. On the contrary, $WIND^d$ and SS^d have positive coefficients, suggesting that these weather variables increase market returns when they affect sentiment. Similar explanations hold for the trading activity variables.

This subsection shows that weather-induced sentiment changes significantly affect the market variables. Whereas previous studies investigate the direct effect of weather on the stock market, we separate this effect into two parts: the effect of weather on investor sentiment and the effect of sentiment on the market variables. We show that weather conditions that do not change investor sentiment do not affect the market. In other words, we show that the weather's effect on the market is stronger when changes in weather are accompanied by changes in investor sentiment.

5. Conclusion

As psychological research has long shown, weather greatly influences people's moods and decision-making processes. Extending these findings, behavioral finance researchers have investigated the relationships between weather and market variables. However, previous studies analyze only the direct relationship between the weather and market variables because of the difficulty of estimating investor sentiment in financial markets. To overcome these limitations, this study estimates intraday market sentiment and separately analyzes the effect of weather variables on market sentiment and that of sentiment on market variables. We show that weather variables, including temperature, humidity, wind, cloud cover, sunshine duration, and rain, significantly affect intraday market sentiment. The weather variables have no significant effect on returns, but they do have significant effects on three trading activity variables: turnover, volatility, and illiquidity. In periods when the weather affects investor sentiment, however, the weather has a stronger effect on the market variables, including returns.

This study is meaningful because it directly shows the relationship between weather and investor sentiment. In other words, our analysis suggests the linkage among weather, investor sentiment,

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and the financial markets, and presents investor sentiment as the intermediate path through which weather affects market variables. This study is also the first to estimate the effects of weather on intraday market variables in the KOSPI market.

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Figure 1. Weekly averages of the weather variables

This figure presents the calendar-week average of each weather variable between 9:00 and 16:00 from June 2000 to December 2021. The left panel of the figure shows trends in *TEMP*, *CC*, and *SS*, and the right panel shows trends in *HUM* and *WIND*.

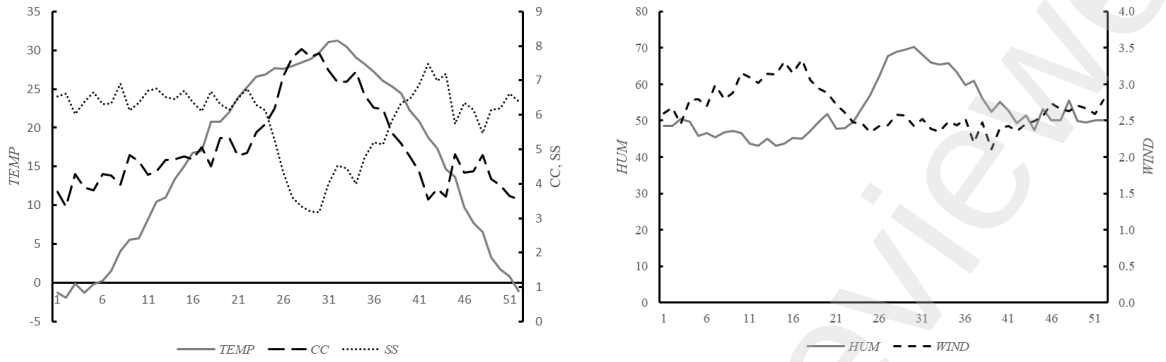

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Table 1. Summary statistics

This table reports summary statistics for the weather, sentiment, and market variables in this study. The variables are observed hourly between 9:00 and 16:00 from June 2000 to December 2021. Panel A shows statistics for the raw weather variables. *TEMP* is the apparent temperature, calculated as $TEMP = 13.12 + 0.6215T - 11.37V^{0.16} + 0.3965V^{0.16}T$, where *T* denotes the temperature in degrees Celsius and *V* denotes the wind speed. *HUM* is the relative humidity in percentage terms, and *WIND* refers to the wind speed. *CC* denotes the cloud cover ratio, which ranges from zero to ten and is defined as the part of the sky covered with clouds relative to the whole sky. *SS* is the sunshine duration and is defined as the percentage of an hour during which direct sunlight reaches the surface of the geographical area without being blocked by clouds or fog. *RAIN* is a dummy variable that equals one if the amount of precipitation is positive and zero otherwise. $RAIN_{EXT}$ refers to extreme rain and equals one if the precipitation level falls in the top tertile. CC_{EXT} and SS_{EXT} denote extreme clouds and extreme sunshine, respectively. CC_{EXT} equals one if the *CC* is ten and zero otherwise, and SS_{EXT} equals one if *SS* equals one and zero otherwise. Panel B shows the summary statistics for the deseasonalized weather variables. We control for seasonality by subtracting the calendar-week average of each weather variable from each raw variable. The sample of weather variables includes non-business days, such as weekends and holidays. Panel C shows the summary statistics for the sentiment and market variables. We calculate the per-minute sentiment using Eq. (6) and then calculate the average per-minute sentiment over an hour. *RET* denotes the logarithm of hourly returns in percentage terms. *TURN, VOL,* and *ILLIQ* refer to turnover, volatility, and illiquidity, respectively, and are mean-adjusted variables.

	MEAN	STD	MIN	Q1	MED	Q3	MAX	N
Panel A. Raw Weather Variables								
TEMP	14.85	10.98	-17.90	5.60	16.50	24.30	39.30	55,034
HUM	52.96	19.82	9.00	37.00	51.00	66.00	100.00	55,034
WIND	2.67	1.34	0.00	1.70	2.50	3.40	11.40	55,034
CC	5.12	3.89	0.00	1.00	6.00	9.00	10.00	55,034
SS	0.59	0.44	0.00	0.00	0.80	1.00	1.00	55,034
RAIN	0.08	0.28						55,034
$RAIN_{EXT}$	0.03	0.17						55,034
CC_{EXT}	0.21	0.41						55,034
SS_{EXT}	0.43	0.50						55,034
Panel B. Deseasonalized Weather Variables								
$TEMP^d$	0.00	4.21	-17.70	-2.49	0.30	2.75	18.03	55,034
HUM^d	0.00	17.15	-48.14	-12.60	-2.67	9.95	62.12	55,034
$WIND^d$	0.00	1.26	-3.71	-0.89	-0.16	0.71	8.55	55,034
CC^d	0.00	3.67	-8.16	-3.68	0.11	3.15	6.88	55,034
SS^d	0.00	4.19	-8.11	-4.12	2.04	3.43	7.70	55,034
Panel C. Sentiment and Market Variables								
Sentiment	0.002	0.710	-6.479	-0.405	0.010	0.413	8.078	33,249
RET(%)	-0.001	0.424	-4.599	-0.181	0.009	0.194	4.590	33,249
TURN	1.000	0.758	0.131	0.520	0.794	1.231	15.873	33,249
VOL	1.000	0.786	0.092	0.499	0.788	1.244	16.963	33,249
ILLIQ	1.000	1.352	0.000	0.222	0.536	1.197	21.774	33,249

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Table 2. Weather and market sentiment: Sorting test

This table reports the results of the sorting test for weather and market sentiment. For each weather variable, the first row, labeled *Average*, shows the average sentiment for each group, and the second row, labeled *Pr(>Med)*, shows the percentage of high sentiment values, that is, sentiment values above the median value, in each group. The columns labeled *Low* and *High* show the average sentiment values (or percentages of high sentiment values) within the lowest and highest weather groups, respectively. The column labeled *Diff* shows the *t*-value (*z*-value) of the difference in the average sentiment values (or percentages of high sentiment values) of the highest and lowest weather groups. For the dummy variables, we calculate the average sentiment value (percentage of high sentiment values) when the variable equals zero and when it equals one and compare the two averages (percentages). The columns labeled *Summer* and *Winter* show results for summer (March to August) and winter (October to February), and we report only the *t*-value or *z*-value of the difference between the lowest and highest groups. The superscripts *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

		Low	2	3	4	High	Diff	Summer	Winter
$TEMP^{d}$	*Average*	0.001	-0.004	0.004	0.016	-0.008	(-0.75)	(-2.00)**	(0.76)
	Pr(>Med)	0.507	0.500	0.500	0.501	0.491	(-1.86)*	(-2.52)**	(-0.21)
HUM^{d}	*Average*	0.012	0.013	0.004	-0.003	-0.016	(-2.30)**	(-1.94)*	(-1.31)
	Pr(>Med)	0.504	0.511	0.499	0.491	0.494	(-1.10)	(-0.74)	(-0.82)
$WIND^{d}$	*Average*	-0.016	-0.002	-0.003	0.013	0.016	(2.58)***	(3.10)***	(0.52)
	Pr(>Med)	0.481	0.498	0.501	0.512	0.506	(2.90)***	(3.81)***	(0.26)
CC^{d}	*Average*	0.009	0.004	-0.003	0.005	-0.007	(-1.28)	(-1.85)*	(0.08)
	Pr(>Med)	0.506	0.494	0.500	0.503	0.495	(-1.28)	(-1.89)*	(0.23)
SS^{d}	*Average*	-0.004	-0.009	-0.002	0.000	0.023	(2.12)**	(2.56)**	(0.60)
	Pr(>Med)	0.495	0.495	0.495	0.501	0.512	(1.99)**	(2.35)**	(0.43)
RAIN	*Average*	0.004				-0.019	(-1.58)	(-1.29)	(-0.99)
	Pr(>Med)	0.501				0.489	(-1.13)	(-0.42)	(-1.41)
$RAIN_{EXT}$	*Average*	0.003				-0.038	(-1.75)*	(-0.95)	(-1.85)*
	Pr(>Med)	0.500				0.484	(-1.00)	(-0.32)	(-1.58)
CC_{EXT}	*Average*	0.000				0.007	(0.62)	(-0.01)	(0.82)
	Pr(>Med)	0.498				0.506	(1.17)	(0.29)	(1.09)
SS_{EXT}	*Average*	-0.006				0.011	(2.17)**	(2.22)**	(1.01)
	Pr(>Med)	0.496				0.504	(1.42)	(1.64)	(0.71)

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Table 3. Weather and market sentiment: Market opening versus the rest of the day

This table reports the results of a sorting test for weather and market sentiment. We divide the full trading day into the market opening period (9:00 – 10:00) (market open) and the rest of the day (10:00 – 15:20). For each weather variable, the first row, labeled *Average*, shows the average sentiment for each group, and the second row, labeled *Pr(>Med)*, shows the percentage of high sentiment values, that is, sentiment values above the median value for each group. The columns labeled *Low* and *High* show average sentiment values (or percentages of high sentiment values) within the lowest and highest weather groups, respectively. The column labeled *Diff* shows the *t*-value (*z*-value) of the difference of the average sentiment values (percentages of high sentiment values) of the highest and lowest weather groups. For the dummy variables, we calculate the average sentiment value (percentage of high sentiment values) when the variable equals zero and when it equals one and compare the two values (percentages). The columns labeled *Market Opening* show the results for the period from 9:00 to 10:00, and the columns labeled *Rest of the Day* show the results for the period from 10:00 to 15:20. The superscripts *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

		Market Opening			Rest of the Day		
		Low	High	Diff	Low	High	Diff
$TEMP^{d}$	*Average*	-0.034	-0.015	(0.42)	0.007	-0.007	(-1.10)
	Pr(>Med)	0.485	0.484	(-0.03)	0.511	0.492	(-2.00)**
HUM^{d}	*Average*	0.051	-0.071	(-3.08)***	0.005	-0.006	(-0.88)
	Pr(>Med)	0.515	0.466	(-2.21)**	0.502	0.499	(-0.24)
$WIND^{d}$	*Average*	-0.029	0.024	(1.25)	-0.014	0.014	(2.25)**
	Pr(>Med)	0.470	0.498	(1.23)	0.482	0.507	(2.65)***
CC^{d}	*Average*	0.021	-0.043	(-1.68)*	0.007	0.001	(-0.42)
	Pr(>Med)	0.497	0.467	(-1.49)	0.508	0.501	(-0.72)
SS^{d}	*Average*	0.051	-0.071	(-3.08)***	0.005	-0.006	(-0.88)
	Pr(>Med)	0.515	0.466	(-2.21)**	0.502	0.499	(-0.24)
RAIN	*Average*	-0.012	-0.003	(0.19)	0.006	-0.022	(-1.96)*
	Pr(>Med)	0.490	0.475	(-0.63)	0.503	0.493	(-0.93)
$RAIN_{EXT}$	*Average*	-0.012	0.021	(0.46)	0.006	-0.050	(-2.35)**
	Pr(>Med)	0.489	0.485	(-0.09)	0.502	0.484	(-1.04)
CC_{EXT}	*Average*	-0.016	0.008	(0.74)	0.004	0.006	(0.28)
	Pr(>Med)	0.488	0.492	(0.25)	0.500	0.509	(1.19)
SS_{EXT}	*Average*	-0.029	0.029	(2.15)**	0.000	0.009	(1.14)
	Pr(>Med)	0.484	0.499	(1.00)	0.499	0.505	(0.93)

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Table 4. Weather and market sentiment: Regression analysis

This table reports the results of regression analyses of the relationship between weather and market sentiment. Using Eq. (7), we regress investor sentiment on each weather variable. The regressions include day-of-the-week dummies, month-of-the-year dummies, intraday trading interval dummies, and lags in sentiment as control variables. We use the rank of each variable instead of the raw sentiment and weather variables. We also estimate the logit regression presented in Eq. (8). We report only the coefficients of each weather variable owing to space limitations. We estimate each rank and logit regression for the entire trading day, and we also estimate separate regressions for the market opening period and the rest of the day. The first three columns, labeled *Rank Regression*, show the results of the rank regressions, and the last three columns, labeled *Logit Regression*, show the results of the logit regressions. The columns labeled *All* show the results for the full trading day, the columns labeled *Open* show the results for the market opening period, and the columns labeled *Other* show the results for the rest of the day. The superscripts *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	Rank Regression			Logit Regression		
	All	Open	Other	All	Open	Other
$TEMP^{d}$	-0.008	0.002	-0.010*	-0.001*	0.000	-0.001*
	(-1.52)	(0.13)	(-1.72)	(3.48)	(0.02)	(3.67)
HUM^{d}	-0.011**	-0.039**	-0.006	-0.001	-0.002**	0.000
	(-1.98)	(-2.45)	(-1.00)	(2.55)	(3.96)	(0.80)
$WIND^{d}$	0.013**	0.012	0.013**	0.001***	0.001	0.001**
	(2.35)	(0.69)	(2.27)	(7.57)	(1.36)	(6.15)
CC^{d}	-0.002	-0.022	0.002	0.000	-0.001	0.000
	(-0.45)	(-1.52)	(0.33)	(0.22)	(1.12)	(0.00)
SS^{d}	0.010*	0.029**	0.006	0.001*	0.001	0.001
	(1.88)	(2.01)	(1.04)	(3.42)	(2.70)	(1.68)
$RAIN$	-0.769	-0.634	-0.773	-0.036	-0.059	-0.030
	(-1.35)	(-0.40)	(-1.28)	(0.79)	(0.38)	(0.44)
$RAIN_{EXT}$	-1.366	0.767	-1.773*	-0.044	-0.005	-0.049
	(-1.46)	(0.30)	(-1.79)	(0.44)	(0.00)	(0.45)
CC_{EXT}	0.435	0.511	0.408	0.036	0.022	0.038
	(1.13)	(0.47)	(1.00)	(1.73)	(0.11)	(1.63)
SS_{EXT}	0.551*	1.721*	0.377	0.025	0.062	0.020
	(1.73)	(1.74)	(1.13)	(1.21)	(1.05)	(0.69)

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Table 5. Weather and market variables: Sorting test

This table reports the results of a sorting test for the weather and market variables. The market variables are returns (*RET*), turnover (*TURN*), volatility (*VOL*), and illiquidity (*ILLIQ*). For each weather variable, the first row, labeled *Average*, shows the average of the market variable of each group, and the second row, labeled *Pr(>Med)*, shows the percentage of the market variable observations that are above the median within each group. We present the *t*-values (*z*-values) of the differences between the average market variables (percentages of high market variables) in the highest and lowest weather groups. For the dummy variables, we compare the difference between the average market variable or percentage of high market variables when the dummy variable equals zero and when it equals one. The superscripts *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

		RET	*TURN*	*VOL*	*ILLIQ*
$TEMP^{d}$	*Average*	(-0.09)	(-5.77)***	(-2.11)**	(-3.16)***
	Pr(>Med)	(-1.03)	(-7.39)***	(-4.21)***	(-2.03)**
HUM^{d}	*Average*	(-0.37)	(7.99)***	(5.15)***	(4.50)***
	Pr(>Med)	(-0.59)	(8.92)***	(6.17)***	(1.93)*
$WIND^{d}$	*Average*	(-0.81)	(-4.99)***	(-5.84)***	(-10.07)***
	Pr(>Med)	(-0.08)	(-2.99)***	(-4.56)***	(-7.02)***
CC^{d}	*Average*	(-1.58)	(0.19)	(0.20)	(1.23)
	Pr(>Med)	(-2.23)**	(1.37)	(-0.03)	(1.43)
SS^{d}	*Average*	(-1.54)	(-0.47)	(-3.67)***	(-10.42)***
	Pr(>Med)	(0.24)	(0.02)	(-1.31)	(-7.23)***
RAIN	*Average*	(1.38)	(3.18)***	(2.15)**	(1.91)*
	Pr(>Med)	(0.60)	(3.23)***	(3.03)***	(1.79)*
$RAIN_{EXT}$	*Average*	(0.44)	(3.42)***	(2.36)**	(3.03)***
	Pr(>Med)	(0.82)	(3.76)***	(4.21)***	(1.80)*
CC_{EXT}	*Average*	(-0.22)	(7.45)***	(3.89)***	(5.28)***
	Pr(>Med)	(-0.90)	(6.53)***	(5.83)***	(5.54)***
SS_{EXT}	*Average*	(0.07)	(-22.69)***	(-16.42)***	(-11.90)***
	Pr(>Med)	(0.55)	(-18.93)***	(-16.48)***	(-9.23)***

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Table 6. Weather and market variables: Regression analysis

This table reports the results of the regression analyses estimating the relationships between the weather and market variables. We replace sentiment (S_t) in Eq. (7) with each market variable and regress each market variable on each weather variable. The regressions include day-of-the-week, month-of-the-year, and intraday trading interval dummies and lags in the market variables as control variables. We use the ranks of each variable instead of the raw variables. We also estimate the logit regression given by Eq. (8), replacing sentiment (S_t) with each market variable. We report the coefficients of each weather variable. The superscripts *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

	RET	*TURN*	*VOL*	*ILLIQ*
$TEMP^d$	-0.007	-0.012***	-0.019***	-0.018***
	(-1.24)	(-5.04)	(-4.73)	(-3.45)
HUM^d	0.004	0.008***	0.013***	0.010*
	(0.69)	(3.24)	(3.37)	(1.89)
$WIND^d$	-0.004	0.001	-0.010**	-0.027***
	(-0.76)	(0.49)	(-2.57)	(-5.36)
CC^d	-0.009*	0.000	-0.001	0.001
	(-1.70)	(0.14)	(-0.16)	(0.22)
SS^d	-0.001	-0.010***	-0.032***	-0.042***
	(-0.26)	(-4.28)	(-8.15)	(-8.20)
RAIN	1.076*	0.223	0.377	0.577
	(1.88)	(0.92)	(0.91)	(1.09)
$RAIN_{EXT}$	1.260	0.612	1.730**	1.701*
	(1.34)	(1.54)	(2.55)	(1.95)
CC_{EXT}	0.275	0.500***	1.066***	1.798***
	(0.71)	(3.04)	(3.81)	(5.00)
SS_{EXT}	-0.515	-0.633***	-1.775***	-2.272***
	(-1.61)	(-4.65)	(-7.66)	(-7.64)

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Table 7. Sentiment and market variables

This table shows the differences in the average market variables according to sentiment levels (Panel A) and the coefficients of sentiment in the regression analyses (Panel B). We divide the full sample into two groups: periods with positive sentiment values and periods with negative sentiment values. We then compare the average *RET*, *TURN*, *VOL*, and *ILLIQ* values for each group. The rows labeled *Positive* and *Negative* show the averages of each market variable in the periods with positive and negative sentiment, respectively, and the row labeled *Diff* shows the differences between the averages. We regress each market variable on sentiment. The regressions include day-of-the-week, month-of-the-year, and intraday trading interval dummies and lags of the market variables as control variables. We use the rank of each variable instead of the raw value. We report the coefficients of sentiment, and *t*-statistics are given in brackets. The superscripts ** and *** denote statistical significance at the 5% and 1% levels, respectively.

Panel A. Difference Test				
Sentiment	*RET*	*TURN*	*VOL*	*ILLIQ*
Positive	0.036	0.981	0.988	0.970
Negative	-0.047	1.030	1.007	1.011
Diff	0.083 ***	-0.049 ***	-0.019 **	-0.041 ***
	(17.72)	(-5.68)	(-2.19)	(-2.71)
Panel B. Regression Test				
	RET	*TURN*	*VOL*	*ILLIQ*
Sentiment	0.236 ***	-0.024 ***	-0.025 ***	-0.008
	(43.56)	(-7.50)	(-5.92)	(-1.48)

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Table 8. Weather, sentiment, and market variables: Sorting test

This table reports the results of the sorting test for only *True* periods. *True* periods are defined as periods in which the weather affects investor sentiment in the expected direction. According to this classification, periods of high (low) sentiment following low (high) values of $TEMP^d$, HUM^d, and CC^d are *True* periods, and periods of high (low) sentiment following high (low) values of $WIND^d$ and SS^d are also *True* periods. Thus, for each weather variable, we divide the sample period into five groups according to the weather values and then separate each group into two subgroups according to the sign of sentiment. We retain only the *True* periods and conduct a sorting test. We present the differences in the average market variables for the highest and lowest weather groups. *T*-statistics are given in brackets, and the superscripts ** and *** denote statistical significance at the 5% and 1% levels, respectively.

	RET	*TURN*	*VOL*	*ILLIQ*
$TEMP^d$	-0.082***	-0.021	0.004	-0.025
	(-8.12)	(-1.18)	(0.23)	(-0.78)
HUM^d	-0.083***	0.137***	0.105***	0.161***
	(-7.90)	(7.20)	(5.34)	(5.02)
$WIND^d$	0.070***	-0.105***	-0.093***	-0.274***
	(6.75)	(-5.52)	(-4.88)	(-7.86)
CC^d	-0.083***	0.039**	0.022	0.083**
	(-7.95)	(2.07)	(1.13)	(2.49)
SS^d	0.064***	-0.026	-0.055***	-0.253***
	(6.20)	(-1.42)	(-2.82)	(-7.73)

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Table 9. Weather, sentiment, and market variables: Regression analysis

This table reports the results of running the regression analysis for only observations in *True* periods. *True* periods are defined as periods in which the weather affects investor sentiment in the expected direction. According to this classification, periods of high (low) sentiment following low (high) values of $TEMP^d$, HUM^d, and CC^d are the *True* periods, as are periods of high (low) sentiment following high (low) values of $WIND^d$ and SS^d. Thus, for each weather variable, we divide the sample period into five groups according to the values of the weather variable. Then, we separate each group into two subgroups according to the sign of the sentiment. We retain only the *True* periods and conduct the regression analysis. We replace sentiment (S_t) in Eq. (7) with each market variable and regress each market variable on each weather variable. The regressions include day-of-the-week, month-of-the-year, and intraday trading interval dummies and lags in the market variables as control variables. We use the ranks of each variable instead of the raw variables. We report the coefficients for each weather variable, and *t*-statistics are given in brackets. The superscripts ** and *** denote statistical significance at the 5% and 1% levels, respectively.

	RET	*TURN*	*VOL*	*ILLIQ*
$TEMP^d$	-0.006***	-0.027	-0.010	0.006
	(-7.68)	(-0.27)	(-0.09)	(0.02)
HUM^d	-0.002***	0.098***	0.106***	0.185***
	(-7.14)	(3.69)	(3.44)	(3.31)
$WIND^d$	0.017***	-1.059***	-1.102***	-3.881***
	(6.16)	(-3.03)	(-2.84)	(-4.73)
CC^d	-0.008***	0.168	0.187	0.693**
	(-7.32)	(1.22)	(1.20)	(2.26)
SS^d	0.007***	-0.091	-0.286**	-1.247***
	(7.13)	(-0.70)	(-2.02)	(-4.57)

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